

Enhanced Radiation Hardness of High-Hydroxyl Fluorine-Doped Silica Optical Fibers

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Abstract We developed high-hydroxyl (OH) fluorine-doped silica optical fibers and evaluated their radiation response at 1550 nm under γ -ray irradiation. The developed fiber exhibits significantly reduced radiation-induced attenuation (RIA) and improved saturation behavior compared with conventional pure-silica-core and fluorine-doped fibers, demonstrating synergistic benefits of combined fluorine and OH incorporation. ©2026 The Author(s)

Introduction

Silica-based optical fibers are increasingly considered for deployment in radiation environments such as nuclear power plants, space missions, and high-energy physics facilities [1]. However, ionizing radiation induces point defects in the silica network, resulting in radiation-induced attenuation (RIA) that degrades signal transmission and sensing reliability [2]. To suppress RIA, pure-silica-core (PSC) fibers and fluorine-doped fibers have been developed, effectively reducing Ge-related and oxygen-related defect formation. Hollow-core fibers also offer extremely low RIA but are inherently incompatible with material-based sensing techniques such as fiber Bragg gratings (FBGs) and scattering-based distributed sensors. Nevertheless, long-term deployment of fiber-optic sensors in extreme radiation environments such as reactor vessels demands optical fibers with radiation hardness beyond that achievable with conventional radiation-hardened fibers. Meanwhile, hydroxyl (OH) groups have been reported to play a complex role in radiation response, either promoting or suppressing the formation of certain defect precursors depending on the wavelength [2,3]. However, the effects of OH concentration in fluorine-doped silica cores on RIA behavior have not been systematically investigated to date.

In this paper, we develop fluorine-doped silica-core optical fibers with intentionally elevated OH concentrations and systematically evaluate their radiation response at 1550 nm under γ -ray irradiation. By comparing commercial radiation-hardened fibers and the developed high-OH fluorine-doped fibers under identical conditions, we demonstrate that the co-presence of fluorine and hydroxyl groups yields a synergistic enhancement in radiation resistance, establishing OH concentration as an actively tunable design parameter for the development of radiation-hardened fibers.

Fabrication of high-hydroxyl (OH) fluorine-doped silica optical fibers

Three types of optical fiber preform (G6, G8, and G10) were fabricated using the vapor axial deposition (VAD) method, with the fluorine and OH concentrations varied to obtain different material compositions. During preform fabrication, the fluorine doping levels were controlled in the range of 0.3–2.0 mol% as shown in Fig. 1(a), while the OH concentrations were adjusted from 0.6 to 3.4 ppm by regulating the hydration conditions during consolidation as shown in Fig. 1(b).

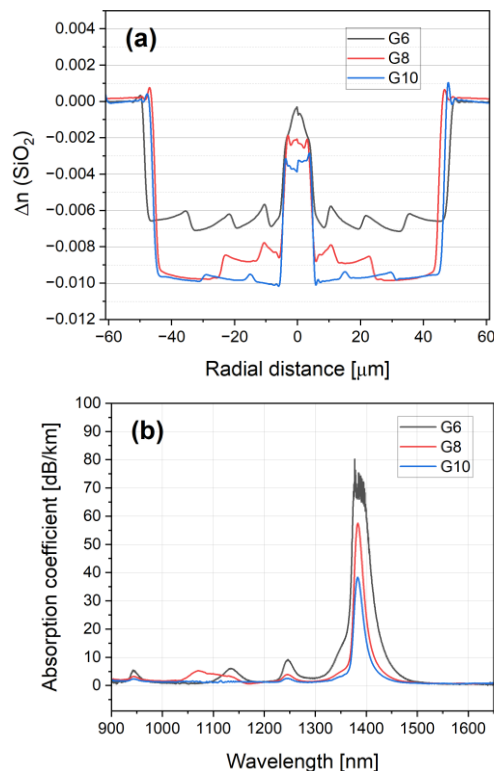


Fig. 1: (a) Refractive index profiles and (b) optical absorption spectra of the fabricated high-OH fluorine-doped fibers. (G6, G8 and G10)

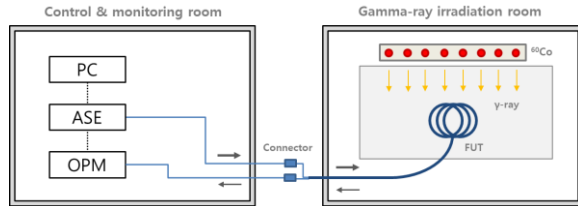
Tab. 1: Optical properties of the optical fibers with different F and OH concentrations

Fiber	Numerical aperture (NA)	Absorption @ 1550 nm	Cut-off wavelength	Fluorine content (core)	OH content (core)
G6 (SMF)	0.123	0.46 dB/km	1174 nm	0.3 mol%	2.1 ppm
G8 (SMF)	0.144	0.53 dB/km	1160 nm	1.6 mol%	0.9 ppm
G10 (SMF)	0.136	0.67 dB/km	1165 nm	2.0 mol%	0.6 ppm

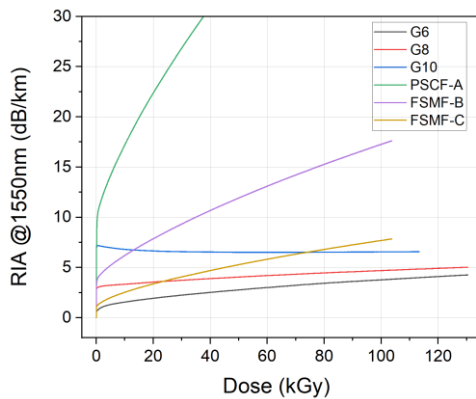
The detailed fiber compositions and optical properties are summarized in Tab. 1. The fluorine and OH concentrations were estimated from the refractive index profiles [4] and the OH absorption peak at 1380 nm [5], respectively.

Radiation-induced attenuation (RIA)

The experimental setup to measure the radiation-induced attenuation (RIA) of optical fiber is shown in Fig. 2. The irradiation was performed at the advanced radiation technology institute (ARTI) under the Korea Atomic Energy Research Institute (KAERI). ^{60}Co was used as a gamma-ray source and a 200 m length (L) of optical fibers was placed inside the irradiation room. Both fiber ends were connected to an amplified spontaneous emission light source (ASE, FiberLabs FL7004) and an optical power meter (OPM, Santec OPM-150), respectively.

**Fig. 2:** Experimental setup to measure the radiation-induced attenuation (RIA) of optical fiber. (FUT: Fiber Under Test)

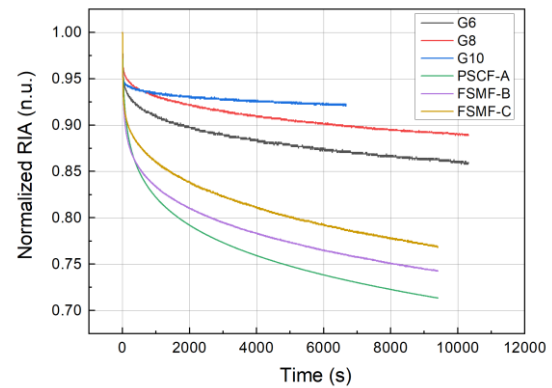
The fibers were irradiated to a total dose exceeding 100 kGy at a dose rate of ~ 6 kGy/h. The RIA was determined from the change in optical output power (P) with respect to radiation dose using Eq. (1). The RIA at 1550 nm of the fabricated fibers was measured together with commercial radiation-hardened fibers for comparison.

**Fig. 3:** RIA at 1550 nm as a function of gamma-ray dose for the fabricated and commercial radiation hardened fibers.

$$\text{RIA} \left[\frac{\text{dB}}{\text{km}} \right] = \frac{P_{\text{initial}}[\text{dBm}] - P_{\text{output}}[\text{dBm}]}{L[\text{km}]} \quad (1)$$

Fig. 3 shows the RIA of the fibers as a function of the exposed gamma-ray dose. For all irradiated fiber samples, the RIA increased sharply after the onset of irradiation, followed by a reduced growth rate or saturation behavior. Among the commercial fibers, PSCF-A exhibited the highest RIA, whereas FSMF-B and FSMF-C, which contain fluorine in the fiber core, showed relatively lower RIA levels with more pronounced saturation behavior compared to PSCF-A. The fibers fabricated in this study (G6, G8, and G10) demonstrated significantly improved RIA performance compared to the commercial fibers. G6 exhibited lower RIA than FSMF-C from the early stage of irradiation. In contrast, G8 and G10 showed a rapid initial increase in RIA. However, the growth rate subsequently decreased significantly, exhibiting strong saturation behavior.

Fig. 4 presents the RIA recovery as a function of time after the end of gamma-ray irradiation. The RIA values were normalized to those measured immediately after irradiation. The RIA recovery occurred immediately after irradiation and was most rapid in the early stage. In general, fibers with relatively higher RIA levels (PSCF-A, FSMF-B, and FSMF-C) also exhibited greater normalized recovery. For example, at 6000 seconds after irradiation, the commercial fibers recovered to normalized levels of 0.73–0.79, whereas the fabricated fibers (G6, G8, and G10) recovered to levels of 0.87–0.93.

**Fig. 4:** RIA recovery at 1550 nm as a function of time after the end of gamma-ray irradiation.

Gamma-ray pre-irradiation effect

When optical fibers are first exposed to high-energy radiation, unstable defect precursors formed during the manufacturing process can be activated, causing high optical losses. Gamma-ray pre-irradiation can reduce the RIA by eliminating these unstable defects prior to deployment in radiation environments. After the first irradiation, the fiber samples were allowed to anneal at room temperature until no significant further RIA recovery was observed. The fibers were then re-irradiated under the same dose rate conditions, and this process was repeated up to four irradiation cycles. The RIA values measured at a dose of 100 kGy for each irradiation cycle are summarized in Fig. 5.

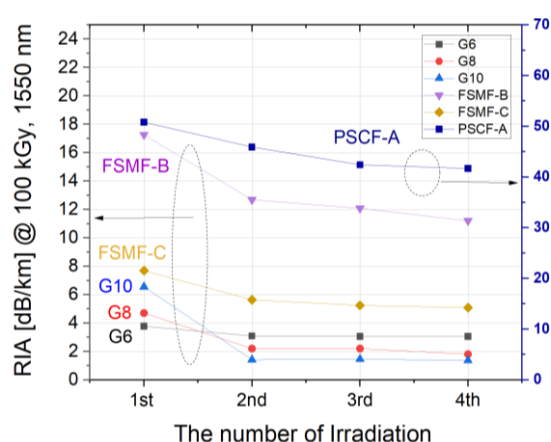


Fig. 5: RIA at 100 kGy of the optical fibers as a function of intermittent gamma-ray pre-irradiation cycles.

Overall, all fibers exhibited a decreasing trend in RIA levels after 100 kGy pre-irradiation treatment. In particular, the RIA after the first pre-irradiation decreased significantly compared to the pristine levels, whereas the additional reduction from the second to fourth pre-irradiation was marginal. Among the tested fibers, PSCF-A and G6, which have no or low fluorine content in the core, showed only a small reduction in RIA through pre-irradiation. In contrast, fibers with higher fluorine content (FSMF-B and FSMF-C) exhibited a more pronounced effect, and fibers containing both fluorine and OH (G8 and G10) exhibited the most significant radiation hardening effect by pre-irradiation. For example, the RIA (dB/km @ 1550 nm) at 100 kGy after the 4th pre-irradiation for PSCF-A and G6 showed RIA reductions of 18% and 19%, respectively, compared to the pristine values. For FSMF-B and FSMF-C, the RIA showed reductions of 35% and 34%, respectively. Notably, G8 and G10 showed reductions from 4.7 dB/km and 6.5 dB/km to 1.8 dB/km and 1.4 dB/km, corresponding to decreases of 62% and 79%, respectively.

Conclusion

In this study, we developed high-OH fluorine-doped silica-core optical fibers via the VAD method and evaluated their radiation response at 1550 nm. The fabricated fibers (G6, G8, and G10) demonstrated lower radiation-induced attenuation (RIA) and improved saturation stability compared to commercial pure-silica-core and fluorine-doped fibers. Specifically, G6, characterized by high OH and low fluorine content, exhibited the highest radiation resistance during the initial exposure up to 100 kGy, highlighting the role of OH in suppressing early-stage defect formation. In contrast, gamma-ray pre-irradiation was most effective for G8 and G10, which contain higher fluorine content, resulting in RIA reductions of 62% and 79%, respectively.

These results suggest a synergistic effect between fluorine and hydroxyl groups in modifying defect formation and stabilization in silica. Overall, this study establishes OH concentration as a key tunable design parameter for developing radiation-hardened optical fibers for harsh environments.

Acknowledgements

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